

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 4 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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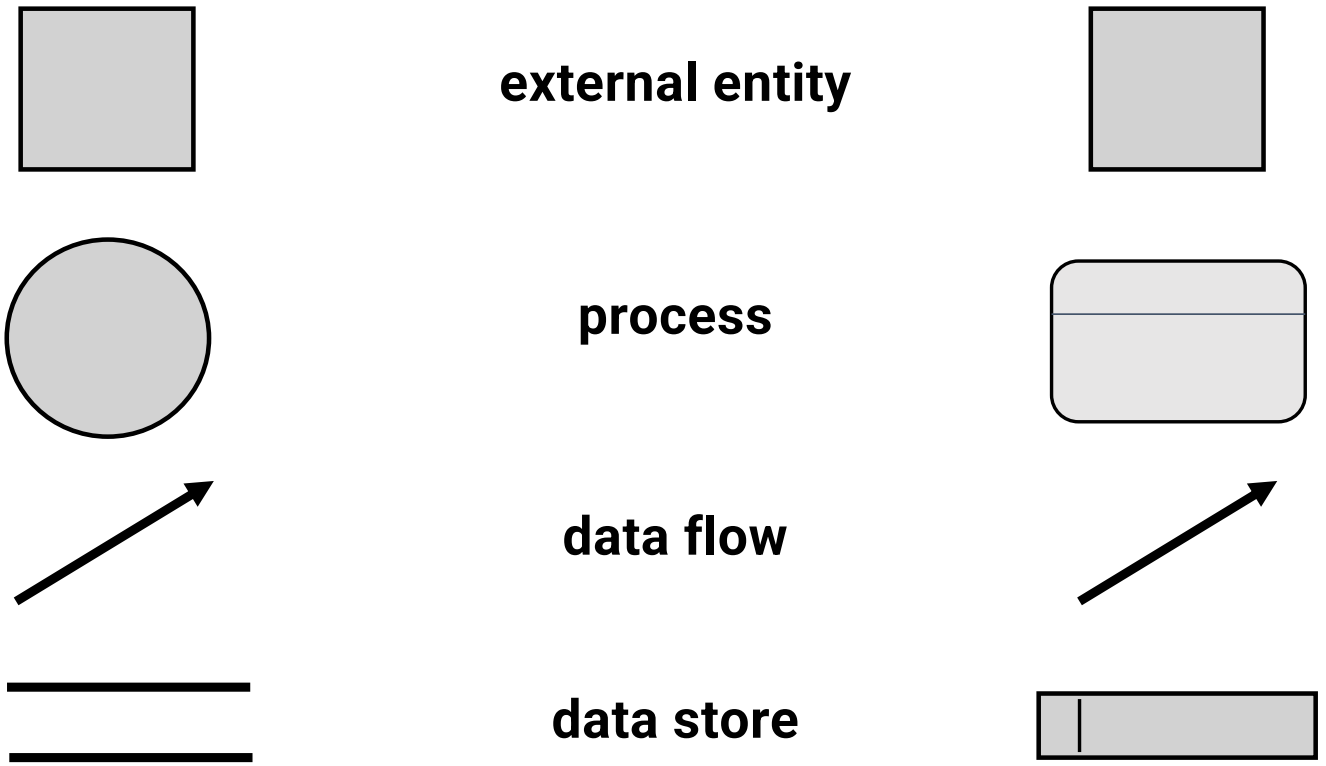
ITDS261 Introduction to Software Engineering

Lecture 7 – Requirements modeling with flow modeling II

Topics

- Recap
  - Flow-oriented modeling
  - Model consistency checking
- Process Specification

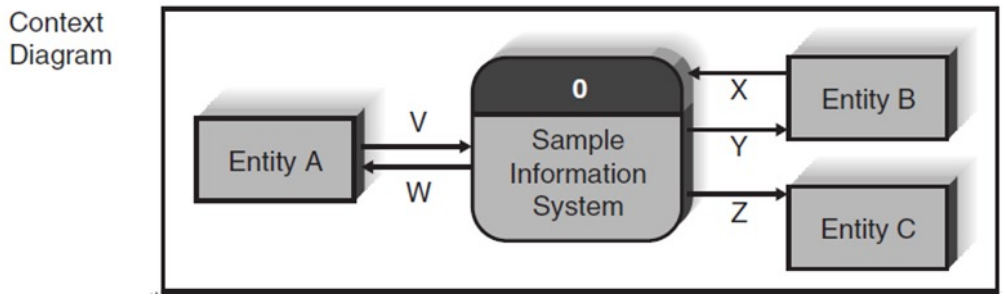
Flow Modeling Notation



Yourdon/DeMarco's

Gane/Sarson's

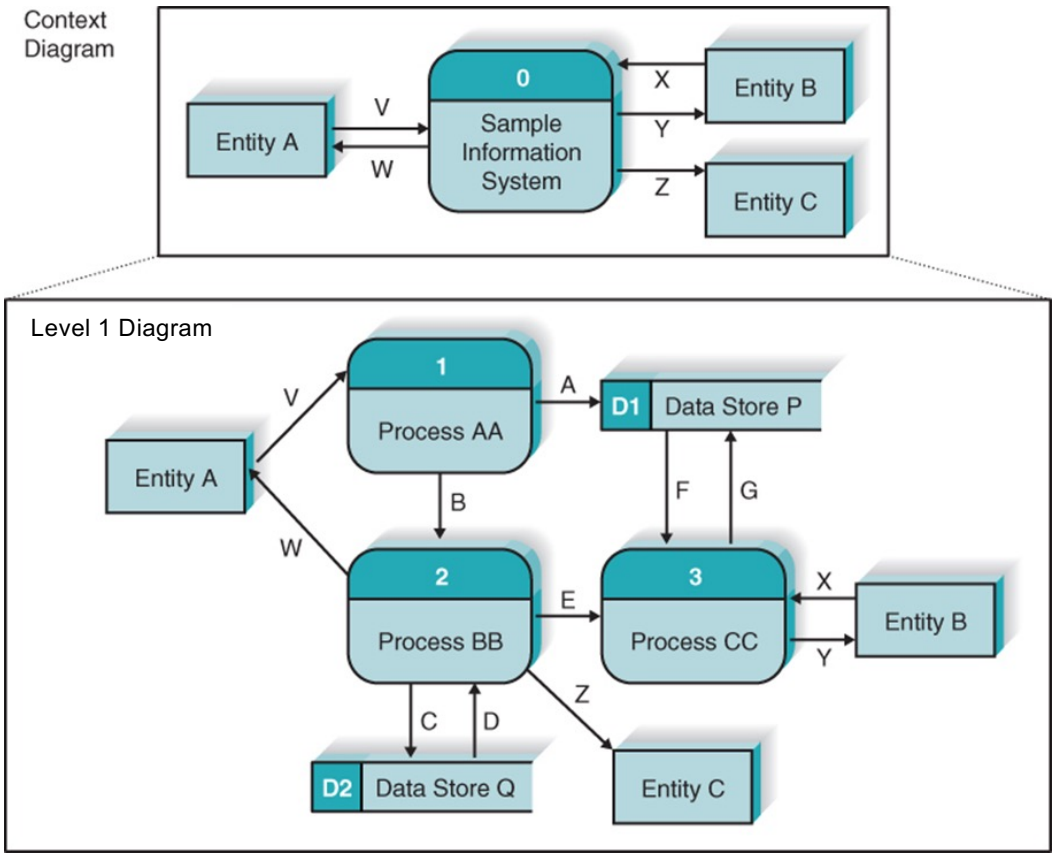
Context Diagram



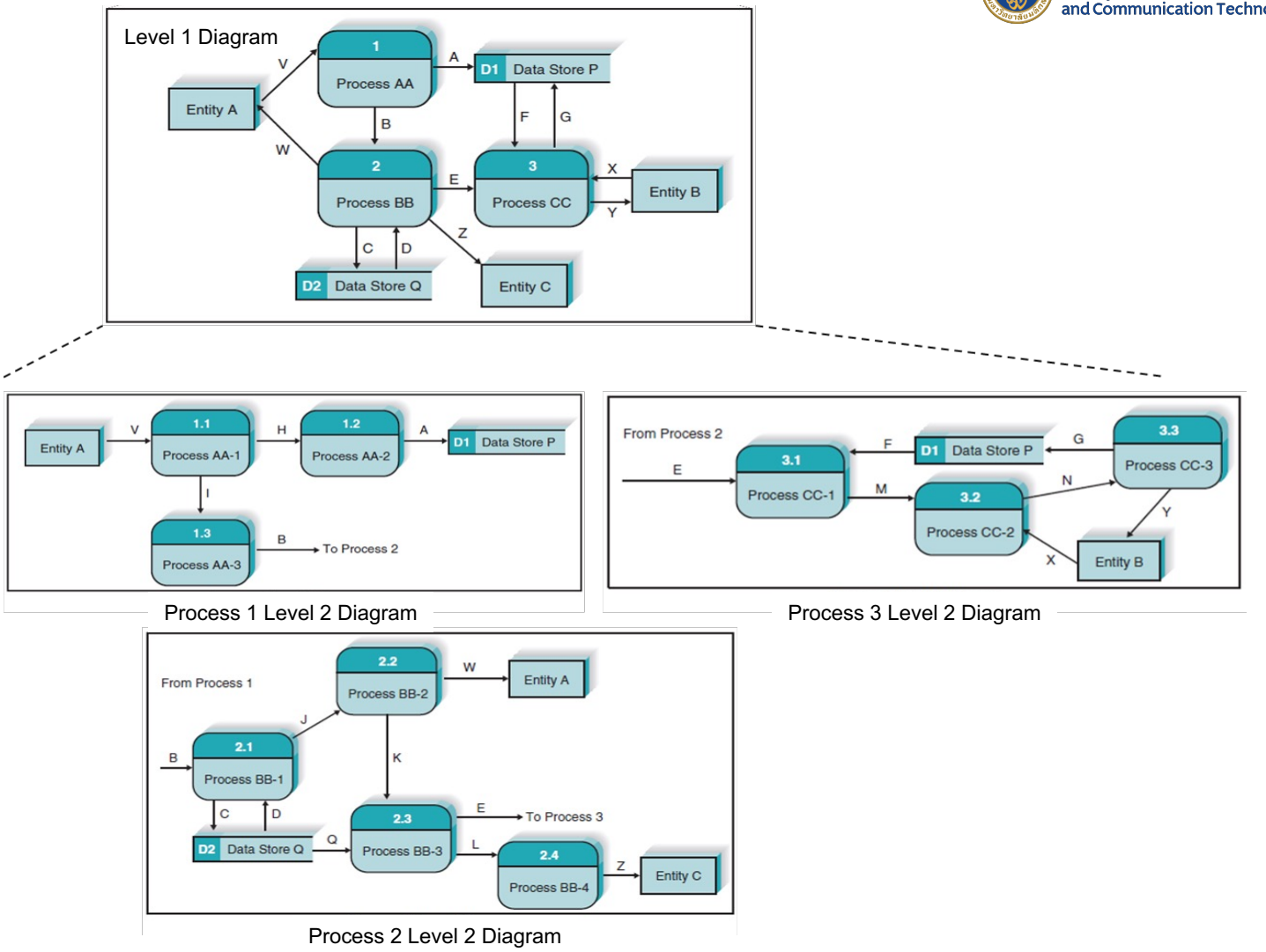
- Top-level DFD in every process model
- Shows the context into which the business process fits
- Shows the overall business process as just one process (process ‘zero’)
- Shows all the external entities that receive information from or contribute information to the system

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The Data Flow Hierarchy



DFD Numbering + Hierarchical Consistency



Process 1 Level 2 Diagram

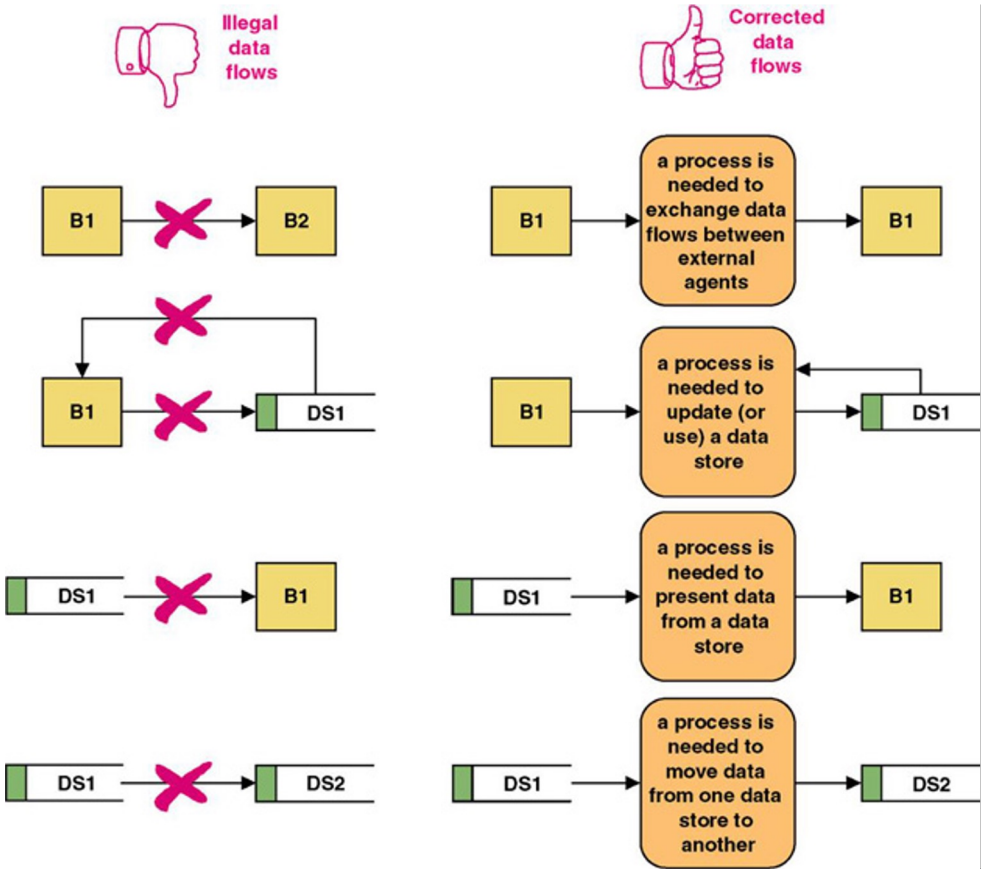
Process 3 Level 2 Diagram

Process 2 Level 2 Diagram

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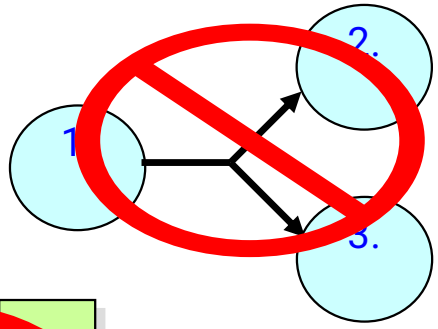
Rules for Data Flows

- A data flow should never go unnamed
- In logical modeling, data flow names should describe the data flow without describing the implementation
- All data flows must begin and/or end at a process

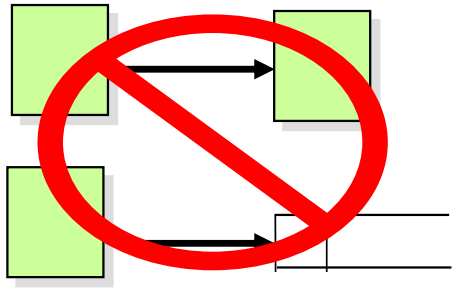


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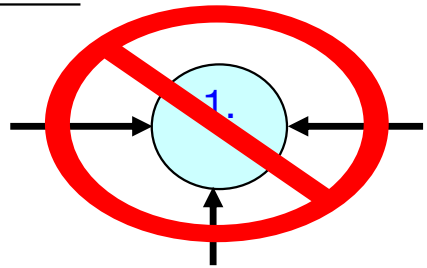
Don’t Break DFD Rules!



Data flows should not split into two or more different data flows



All data flows must EITHER originate or terminate at a process

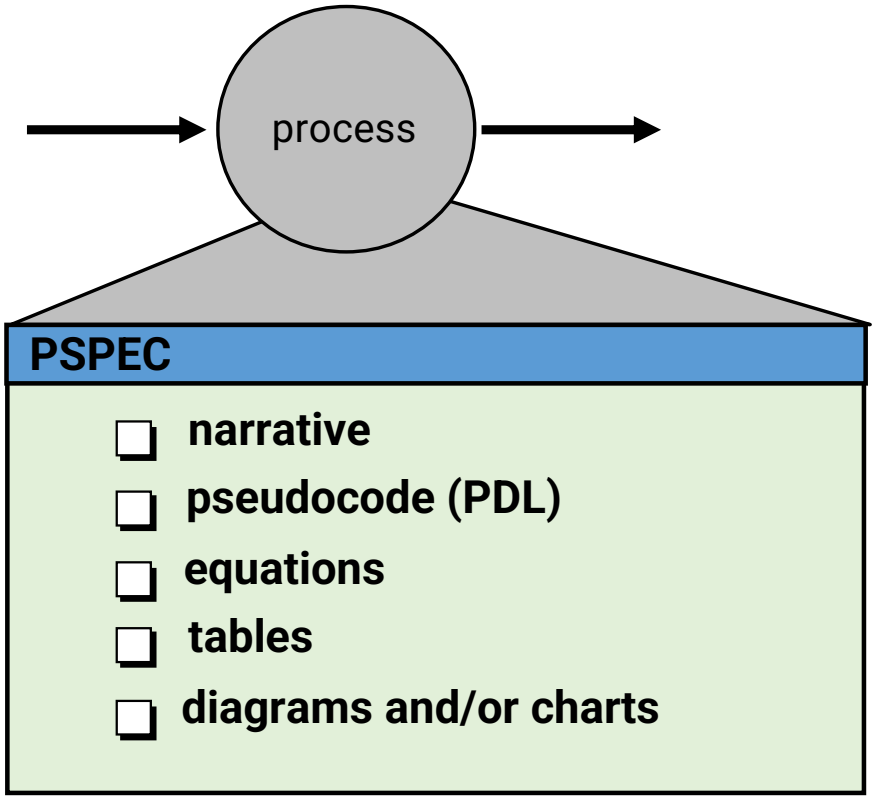


Processes need to have at least one input data flow and one output data flow

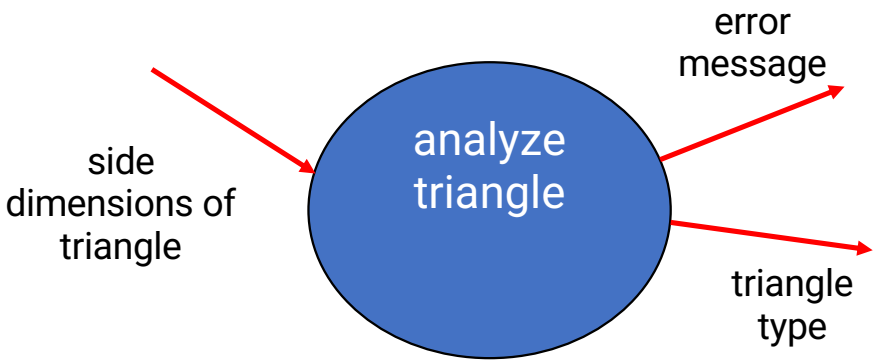
Explaining the Processes

Process Specification / Structured English / Decision Table

Process Specification (PSPEC)



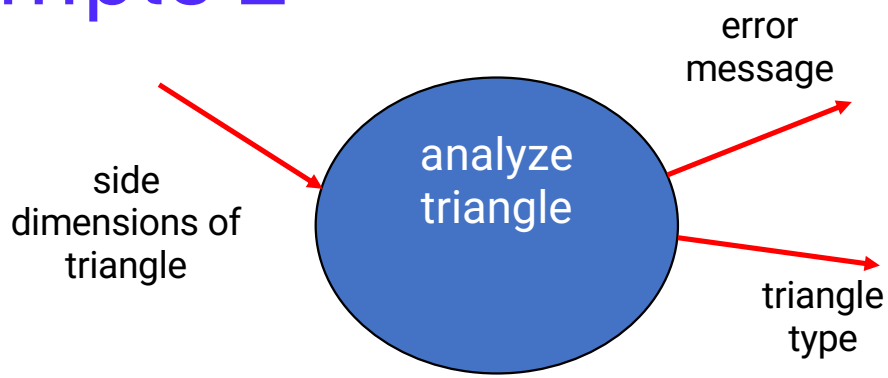
PSPEC Example 1



PSPEC: Processing Narrative for Analyze Triangle

The analyze-triangle process accepts values A, B, and C that represent the side dimensions of a triangle. The process tests the dimension values to determine whether all values are positive. If a negative value is encountered, and error message is produced. The process evaluates valid input data to determine whether the dimensions define a valid triangle and if so, what type of triangle—equilateral, isosceles or scalene– is implied by the dimensions. The type is output.

PSPEC Example 2



PSPEC: PDL for Analyze Triangle

```
read side dimensions;
  if any dimension is negative then produce error message;
  if the largest dimension is less than the sum of the others
    then
       determine number of equal sides;
       if three sides are equal then type is equilateral;
       if two sides are equal then type is isosceles;
       if no sides are equal then 1) type is scalene; 2) output triangle type;
else output type = 0, indication that no triangle exists;
```

Structured English

Structured English – a language syntax for specifying the logic of a process

- Based on the relative strengths of structured programming and natural English

1. For each CUSTOMER NUMBER in the data store CUSTOMERS:
   a. For each LOAN in the data store LOANS that matches the above CUSTOMER NUMBER:
      1) Keep a running total of NUMBER OF LOANS for the CUSTOMER NUMBER.
      2) Keep a running total of ORIGINAL LOAN PRINCIPAL for the CUSTOMER NUMBER.
      3) Keep a running total of CURRENT LOAN BALANCE for the CUSTOMER NUMBER.
      4) Keep a running total of AMOUNTS PAST DUE for the CUSTOMER NUMBER.
   b. If the TOTAL AMOUNTS PAST DUE for the CUSTOMER NUMBER is greater than 100.00 then
      1) Write the CUSTOMER NUMBER and data in the data flow LOANS AT RISK.
   Else
      1) Exclude the CUSTOMER NUMBER and data from the data flow LOANS AT RISK.

Structured English Constructs (Part 1)

Construct	Sample Template
Sequence of steps – Unconditionally perform a sequence of steps.	[Step 1] [Step 2] … [Step n]
Simple condition steps – If the specified condition is true, then perform the first set of steps. Otherwise, perform the second set of steps. Use this construct if the condition has only two possible values. (Note: The second set of conditions is optional.)	**If** [truth condition] **then** [sequence of steps or other conditional steps] **else** [sequence of steps or other conditional steps] ~~**End If**~~
Complex condition steps – Test the value of the condition and perform the appropriate set of steps. Use this construct if the condition has more than two values.	**Do the following based on** [condition]: **Case 1: If** [condition] = [value] then [sequence of steps or other conditional steps] **Case 2: If** [condition] = [value] then [sequence of steps or other conditional steps] … **Case n: If** [condition] = [value] then [sequence of steps or other conditional steps] ~~**End Case**~~

Structured English Constructs (Part 2)

Multiple conditions – Test the value of multiple conditions to determine the correct set of steps.

Use a decision table instead of nested if-then-else Structured English constructs to simplify the presentation of complex logic that involves combinations of conditions.

***A decision table** is a tabular presentation of complex logic in which rows represent conditions and possible actions and columns indicate which combinations of conditions result in specific actions.*

DECISION TABLE	Rule	Rule	Rule	Rule
[Condition]	value	value	value	value
[Condition]	value	value	value	value
[Condition]	value	value	value	value
[Sequence of steps or conditional steps]	X			
[Sequence of steps or conditional steps]		X	X	
[Sequence of steps or conditional steps]				X

Although it isn't a Structured English construct, a decision table can be named, and referenced within a Structured English procedure.

One-to-many iteration – Repeat the set of steps until the condition is false.

Use this construct if the set of steps must be performed at least once, regardless of the condition's initial value.

Repeat the following until [truth condition]:
[sequence of steps or conditional steps]
~~**End Repeat**~~

Zero-to-many iteration – Repeat the set of steps until the condition is false.

Use this construct if the set of steps is conditional based on the condition's initial value.

Do while [truth condition]:
[sequence of steps or conditional steps]
~~**End Do**~~

- OR -

For [truth condition]:
[sequence of steps or conditional steps]
~~**End For**~~

Policies and Decision Tables

Policy – a set of rules that govern show a process is to be completed

Decision table – a tabular form of presentation that specifies a set of conditions and their corresponding actions

- As required to implement a policy

A Simple Decision Table

A SIMPLE POLICY STATEMENT

CHECK CASHING IDENTIFICATION CARD

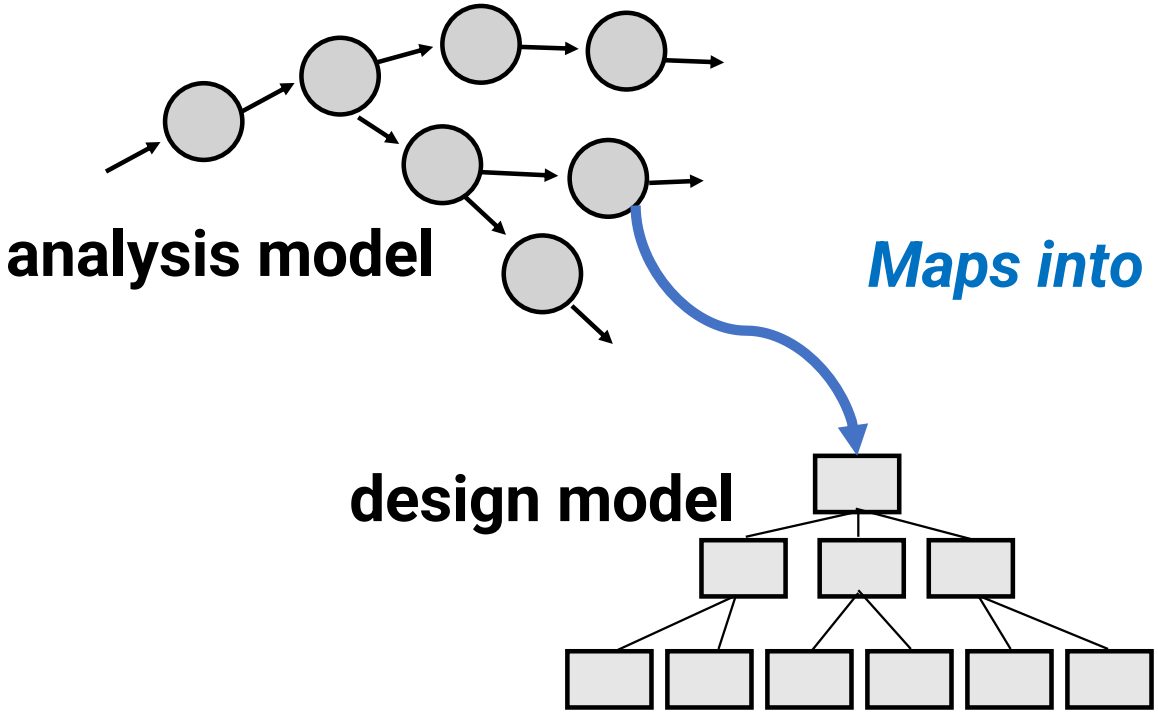
A customer with check cashing privileges is entitled to cash personal checks of up to $75.00 and payroll checks from companies pre-approved by *LMART*. This card is issued in accordance with the terms and conditions of the application and is subject to change without notice. This card is the property of *LMART* and shall be forfeited upon request of *LMART*.

SIGNATURE *Charles C. Parker, Jr.*
EXPIRES **May 31, 2006**

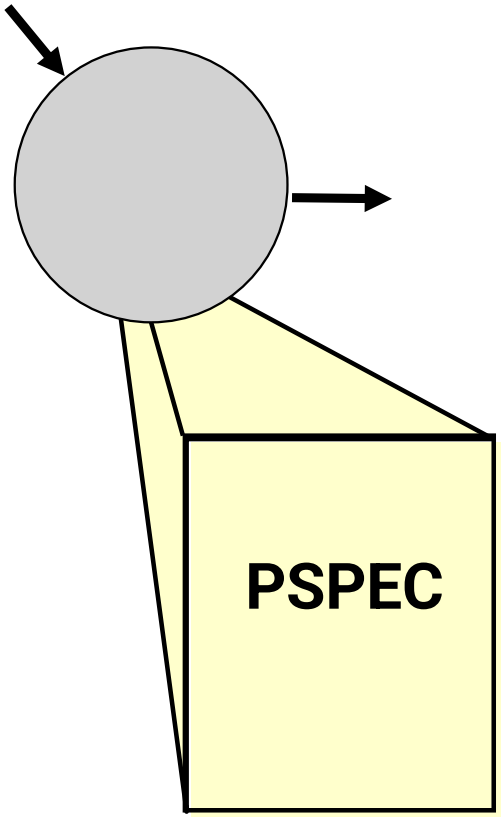
THE EQUIVALENT POLICY DECISION TABLE

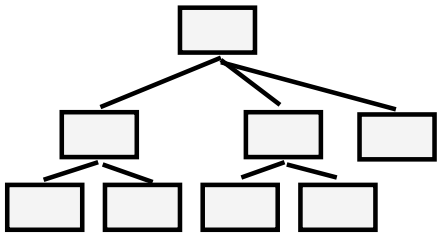
	Conditions and Actions	Rule 1	Rule 2	Rule 3	Rule 4
Condition Stubs	C1: Type of check	personal	payroll	personal	payroll
	C2: Check amount less than or equal to $75.00	yes	doesn't matter	no	doesn't matter
	C3: Company accredited by *LMART*	doesn't matter	yes	doesn't matter	no
Action Stubs	A1: Cash the check	X	X		
	A2: Don't cash the check			X	X
		Rules			

DFDs: A Look Ahead



A Design Note





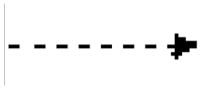
one or more "components" in the software design

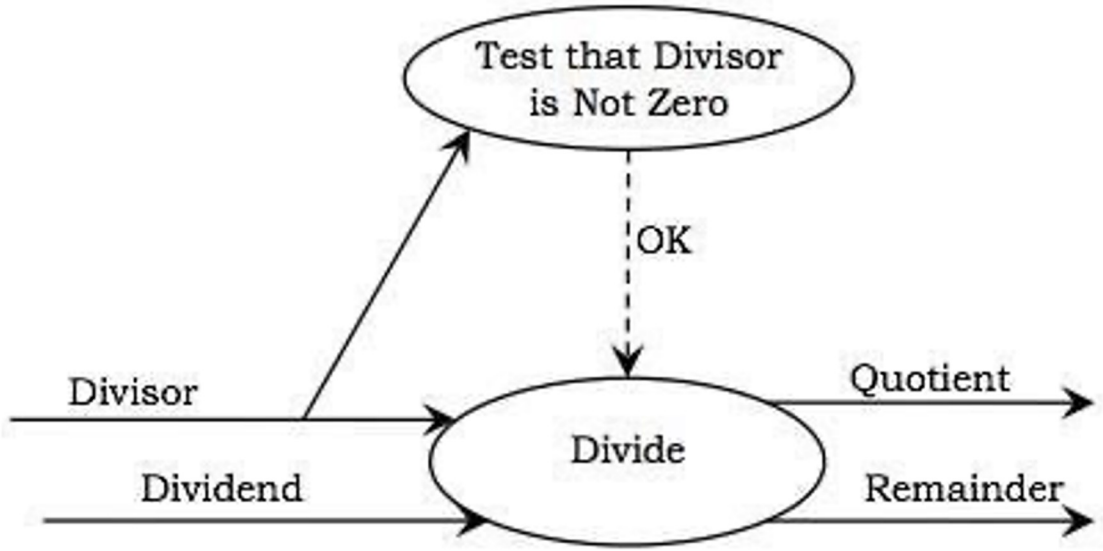
Control Flow Modeling

Control Flow Modeling

- A control flow represents **“events”** and the processes that manage events.
- An “event” is a Boolean condition that can be ascertained by:
  - listing all sensors that are "read" by the software
  - listing all interrupt conditions
  - listing all "switches" that are actuated by an operator
  - listing all data conditions
  - recalling the noun/verb parse that was applied to the processing narrative, review all "control items" as possible CSPEC inputs/outputs
- A control flow is a signal that carries out a command or indicates that something has occurred.
- A control flow occurs at a discrete point in time.

Control Flows in DFD

- Represented as a dotted arrow line 
- The arrow indicates the direction of the control flow.
- The name of the event is written beside the arrow.



Control Specification (CSPEC)

The CSPEC can be:

- state diagram (sequential spec)
- state transition table
- decision tables
- activation tables

combinatorial spec

Exercise

- Use the case of the zoo ticketing system
  - Draw the DFD Level 1 of the system
  - Draw the DFD Level 2 of one process that is defined in your DFD Level 1